Personalized Portfolio Maker

A Web Application for Instant Portfolio Generation

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***Abstract*—The Personalized Portfolio Maker is a free, browser-based web application that enables students and early-career professionals to generate polished, downloadable portfolio websites instantly ,without any coding knowledge, server setup, or paid subscription. The system guides users through a structured input form covering personal details, skills, projects, education, and contact information. A client-side rendering engine transforms this data in real time into a fully styled, self-contained portfolio page, drawn from professionally designed templates spanning professional, creative, and developer aesthetics. Users can further personalize their output by choosing from different accent colour presets, typography options, and layout modes. A live preview updates instantly, and the final portfolio can be exported as a standalone HTML file, a print-ready PDF, or a structured CV/Resume document. By eliminating cost barriers and technical complexity, the platform addresses a clear gap in the existing ecosystem of portfolio tools. Future development will incorporate AI-driven content suggestions to further streamline the creation experience.**

Keywords—Portfolio Generator, Web Application, Template Engine, PDF Export, CV Generator, Client-Side Architecture, Student Tools.

I. Introduction

In today's digital-first employment landscape, a personal portfolio website has become an indispensable asset for students seeking internships and entry-level roles. Unlike a static resume, an online portfolio communicates creativity, technical depth, and professional identity simultaneously. However, the majority of tools available for creating such portfolios , platforms such as Wix, Squarespace, or Webflow ,are subscription-based, impose third-party branding on free tiers, and present complex interfaces that assume prior design or development experience.

For undergraduate students, particularly those with limited time and financial resources, these barriers are significant. Building a portfolio from scratch using raw code demands knowledge of HTML, CSS, JavaScript, and responsive design principles , a combination that many non-specialist students do not yet possess. This gap between the need for a professional online presence and the practical means to create one motivates the development of the Personalized Portfolio Maker.

The proposed system, PortEdge, is a fully browser-based, zero-installation web application. It operates without any server-side backend or database, making it inherently accessible, privacy-preserving, and cost-free. A student fills in a guided step-by-step form, selects a visual template, customizes the colour and typography, previews the result live, and downloads a production-ready portfolio ,all within a single session in any modern web browser.

II. Problem statement

The core challenge this work addresses is the absence of a free, beginner-friendly, and feature-complete platform for student portfolio creation. Specifically, the following problems exist in the current landscape:

- The majority of professional portfolio builders operate on a freemium or subscription model, restricting essential features , such as custom domains, PDF export, and ad-free pages , to paying customers.
- Tools with free tiers prominently display platform branding, undermining the professional appearance of the student's portfolio.
- Platforms such as WordPress require web hosting setup, server-side configuration, and plugin management, all of which demand intermediate technical skills beyond a first- or second-year student's typical capability.
- No-code editors such as Webflow, while powerful, have steep learning curves and are optimised for complex multi-page websites rather than simple, single-page student portfolios.
- Existing free tools do not integrate CV and Resume generation alongside portfolio creation, forcing students to use multiple separate platforms.
- Creating a portfolio from scratch requires expertise in responsive design, typography, and front-end development ,a prerequisite that excludes a large segment of the student population.

III. Objectives

- To develop a free, open, and zero-installation web application that allows any student to generate a professional portfolio without coding knowledge.
- To provide a guided, multi-section input interface covering personal information, skills, projects, education, and contact details.
- To implement a real-time live preview so that users can see exactly how their portfolio will look before downloading it.
- To offer professionally designed templates spanning professional, creative, and developer styles, with customizable colours, fonts, and layouts.
- To support three export formats: a standalone HTML file, a browser-triggered PDF, and a formatted CV/Resume document.
- To persist user data locally across browser sessions, preventing data loss and enabling users to return and edit their portfolio at any time.
- To establish an extensible architecture that can accommodate future enhancements, particularly AI-based content suggestions.

Iv. Literature review

Several prior works address the portfolio creation problem from different angles. Sharma and Pundhir [1] conducted a comparative study of portfolio builder platforms (2024) and identified three recurring weaknesses in existing tools: limited free-tier customization, heavy template dependency, and the absence of personal branding metrics. Nagaraju et al. [2] proposed a Django-based student portfolio system (2025) featuring authentication and dynamic content management, but their architecture introduces internet dependency and server maintenance overhead , limitations that PortEdge eliminates by operating entirely client-side. Pandey et al. [3] (2021) built a Flask-based generator using LinkedIn scraping and a Naive Bayes classifier for career recommendations, acknowledging the value of intelligent suggestions , a direction PortEdge adopts with a curated skill hint list today and a full AI integration planned for future work. Dash et al. [4] (2024) and Priyadharshan et al. [5] (2025) each implemented full-stack portfolio systems with PDF generation, yet both required server infrastructure and noted the absence of AI-based template suggestions. PortEdge surpasses these in feature completeness while requiring no server setup whatsoever.

Table 1. Comparison of Existing Systems vs. PortEdge

Feature	Wix / Squarespace	WordPress	Webflow	PortEdge (Proposed)
Cost	Paid / Freemium	Free + Hosting	Paid	Completely Free
Ease of Use	Medium	Complex	Steep	Simple Wizard
PDF / CV	Not Available	Plugin Only	Not Built-in	Built-In
Templates	500+ (Paid)	1000+ (Paid)	40+ (Paid)	13 Free Templates
Live Preview	Yes	Partial	Yes	Yes (Real-Time)
No-Code	Yes	No	Partial	Yes
Auto-Save	Cloud (Paid)	No	Cloud (Paid)	Browser Storage

V. Methodology

A] Proposed system:

PortEdge is a modular, event-driven Single-Page Application (SPA) that operates entirely within the client's browser. It is built using HTML5 for structure, CSS3 for presentation and theming, and Vanilla JavaScript (ES6+) for all application logic , with no external frameworks or libraries. Google Fonts provides typography for the generated outputs. Browser-native localStorage enables persistent, session-independent data storage, while the Blob API and the browser's built-in print engine handle file downloads and PDF generation respectively.

The user journey follows a sequential five-screen wizard: (1) Landing - introduces the platform and initiates the process; (2) Form - a multi-section input interface covering Personal Information, Skills, Projects, Education, and Contact & Social details; (3) Templates - a grid of portfolio themes spanning Professional, Creative, and Developer categories; (4) Customize - a panel for selecting accent colour, typography, layout mode, and dark-mode preference; and (5) Preview- a live iframe rendering of the complete portfolio with export controls.

All user data is maintained in a singleton in-memory state object. Every field update triggers an immediate write to localStorage as serialized JSON. On page reload, the stored data is deep-merged with the application's default schema, ensuring backward compatibility. This design allows users to close the browser and return to find all previously entered data intact - without any account creation or cloud service.

B] System Architecture

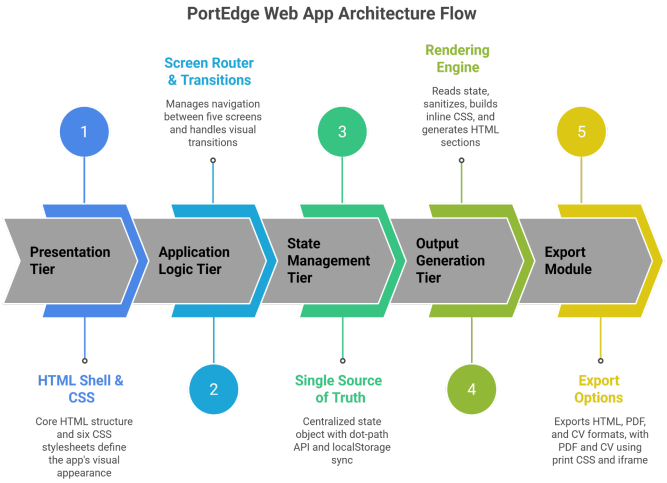

fig 1.System Architecture

PortEdge follows a four-tier client-side architecture: Presentation, Application Logic, State Management, and Output Generation. No network communication occurs during operation , the entire lifecycle runs within the browser.

The Presentation tier comprises the HTML shell and sox CSS stylesheets defining all five screens and their visual styles. The Application Logic tier acts as a screen router, managing transitions, updating a progress indicator, and triggering

screen-specific initialization. The State Management tier owns the single source of truth for all application data, exposing a dot-path API and synchronizing the in-memory object with localStorage on every write. The Output Generation tier consumes the state snapshot through two subsystems: the Rendering Engine and the Export Module.

The Rendering Engine is the technical core of PortEdge. When invoked, it reads the complete application state, sanitizes all user-supplied strings using an HTML entity escaper to prevent XSS vulnerabilities, constructs an inline CSS block embedding the user's chosen accent colour, font stack, layout class, and dark-mode rules, assembles semantic HTML sections (hero header, skills chip grid, project cards with technology tags, education section, and contact row), and wraps the result in a complete self-contained HTML document. The selected template drives a specific CSS class on the portfolio shell, producing the distinct visual character of each theme.

The Export Module provides three pipelines: (1) HTML Export - delivers the rendered document as a downloadable .html file via the Blob API, self-contained and hostable on any static platform; (2) PDF Export - injects print-optimized CSS (A4 dimensions, 10 mm margins, animation suppression), loads the document into the preview iframe, waits for fonts and images to fully load, and triggers the browser's native print dialog; (3) CV/Resume Export - generates a separate structured resume document following a standard two-section format, exported as an HTML file that users print to PDF in any browser.

VI.. Results & Discussion

1. Landing Page:

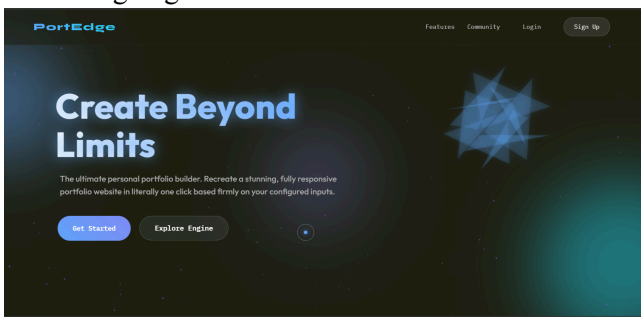

Fig 2.Landing page

The Landing Page serves as the entry point of the application, introducing users to the portfolio builder platform. It features a prominent heading, brief description, and clear call-to-action buttons such as “Get Started” and “Explore Engine.” A top navigation bar provides access to features, community, login, and signup options. Users can interact by navigating through sections or initiating the portfolio creation process. The design uses a modern dark theme with visual elements to enhance engagement and create a strong first impression.

2.Personal Information:

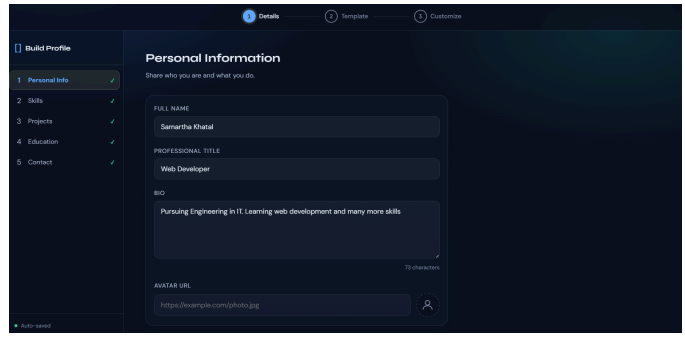

Fig 3.Personal Information

The Personal Information screen allows users to input their basic details required for building a portfolio. It includes form fields such as full name, professional title, bio, and avatar URL. A sidebar navigation shows different sections like skills, projects, education, and contact, along with progress indicators. Users interact by entering their information, which is automatically saved. This screen ensures structured data collection and forms the foundation for generating a personalized portfolio.

3.Add Skills

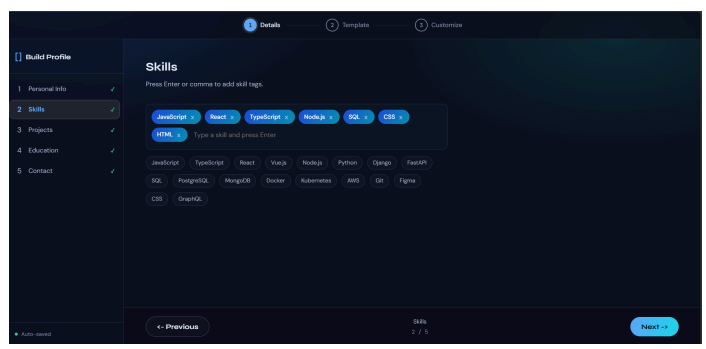

Fig 4.Skills section

The Skills screen enables users to add and manage their technical or professional skills. It provides an input area where users can enter skills using tags, along with suggested skill options for quick selection. Selected skills are displayed as removable tags for easy editing. Navigation buttons allow users to move between steps in the profile-building process. This screen simplifies skill entry and helps in organizing competencies effectively for the final portfolio output.

4.Template Selection page:

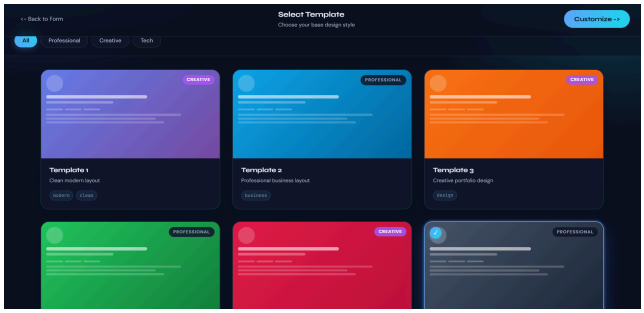

Fig 5.Template Selection Page

The Template Selection screen allows users to choose a base design for their portfolio. It displays multiple template cards categorized under filters such as All, Professional, Creative, and Tech. Each template includes a preview, title, tags, and category labels to help users make an informed choice. Users can interact by browsing, filtering, and selecting a preferred template, which is visually highlighted upon selection. A navigation option to proceed to customization is also provided. This screen simplifies design selection and enhances user control over the portfolio's appearance.

5.Portfolio Preview Section

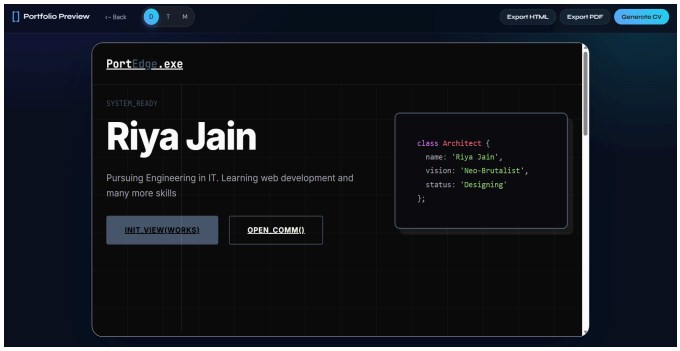

Fig 6. Portfolio Preview Section

The Portfolio Preview screen provides a live representation of the generated portfolio based on user inputs and selected template. It includes options to switch between different device views (desktop, tablet, mobile) and allows users to review the final layout. Action buttons such as "Export HTML," "Export PDF," and "Generate CV" enable users to download or finalize their portfolio. Users can navigate back for edits if needed. This screen ensures users can verify and finalize their portfolio before exporting.

6.Generated CV :

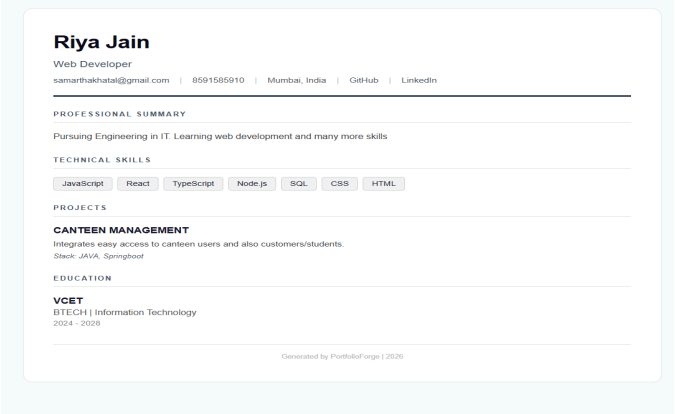

Fig 7. Generated CV

The CV Generation screen presents the final Curriculum Vitae (CV) generated from the user's provided information in a structured and professional format. It organizes details such as personal information, skills, projects, education, and contact into clearly defined sections. Users can review the content before exporting it. The screen provides an option to

download the CV in PDF format, ensuring easy sharing and usage. This screen serves as the final output stage, allowing users to obtain a polished and ready-to-use CV based on their input data.

VII. Conclusion

This paper presented PortEdge, a free, browser-based Personalized Portfolio Maker designed to remove the technical and financial barriers that prevent students from building professional online portfolios. By operating entirely on the client side,PortEdge delivers an accessible and privacy-preserving alternative to subscription-based platforms.

The system provides a complete, end-to-end workflow: a guided multi-section input form, professionally designed templates, full theme customization, a real-time live preview, and three export formats including HTML, PDF, and CV/Resume. It compares favorably against existing tools on cost, ease of use, and feature completeness -particularly for undergraduate students preparing for internship and placement cycles.

Future development will focus on AI-based content generation, cloud persistence, and direct deployment integration, evolving PortEdge from a portfolio generator into a comprehensive personal branding tool for students at every stage of their career.

VIII. Future Scope

•AI-Based Content Suggestions: Integration of a language model API to generate bio descriptions, project summaries, and skill recommendations based on the user's stated career goal.

•Cloud Storage and Cross-Device Sync: A lightweight backend to allow users to save portfolios online and access them from any device.

•One-Click Deployment: Direct integration with platforms such as GitHub Pages or Netlify to publish the portfolio to a live URL from within PortEdge.

•Direct Image Upload: Base64 encoding of profile photos uploaded from the user's device, eliminating the need for external image hosting.

•Recruiter Simulation and Branding Score: An AI module that evaluates portfolio completeness, keyword density, and visual clarity and returns improvement suggestions.

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